

Advanced Calculus HW 2

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Question 1

(a)

To find $\mathcal{D}(\vec{r})$ we need to find the intersections of the natural domains of the component functions.

$$\begin{aligned} \ln(t), \quad -t, \quad t^2 \\ t > 0, \quad t \in \mathbb{R}, \quad t \in \mathbb{R} \\ t > 0 \end{aligned}$$

$$\mathcal{D}(\vec{r}) = t, \quad t > 0$$

(b)

$$\frac{d\vec{r}}{dt} = \frac{1}{t}\hat{i} - 1\hat{j} + \frac{t}{2}\hat{k}, \quad t > 0$$

Yes, this curve is smoothly parameterized since $\vec{r}'(t) \neq 0 \quad \forall t > 0$ since $-1 \neq 0$ and all component functions are continuous for $t > 0$

(c)

$$\begin{aligned} \left| \frac{d\vec{r}}{dt} \right| &= \sqrt{\left(\frac{1}{t} \right)^2 + (-1)^2 + \left(\frac{t}{2} \right)^2} \\ &= \sqrt{\left(\frac{1}{4t^2} \right) (t^4 + 4t^2 + 4)} \\ &= \left(\frac{1}{2t} \right) (t^2 + 2) \end{aligned}$$

(d)

$$\begin{aligned} \vec{T}(t) &= \frac{\vec{r}'(t)}{|\vec{r}'(t)|} \\ &= \left(\frac{2t}{t^2 + 2} \right) \left(\frac{1}{t}\hat{i} - 1\hat{j} + \frac{t}{2}\hat{k} \right) \\ &= \frac{2}{t^2 + 2}\hat{i} - \frac{2t}{t^2 + 2}\hat{j} + \frac{t^2}{t^2 + 2}\hat{k}, \quad t > 0 \end{aligned}$$

(e)

$\ln(t) = 0$ if and only if $t = 1$

$$\vec{T}(1) = \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k}$$

The line, L is then

$$L(\lambda) = \left(0, -1, \frac{1}{4} \right) + \lambda \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right), \quad \lambda \in \mathbb{R}$$

Question 2

(a)

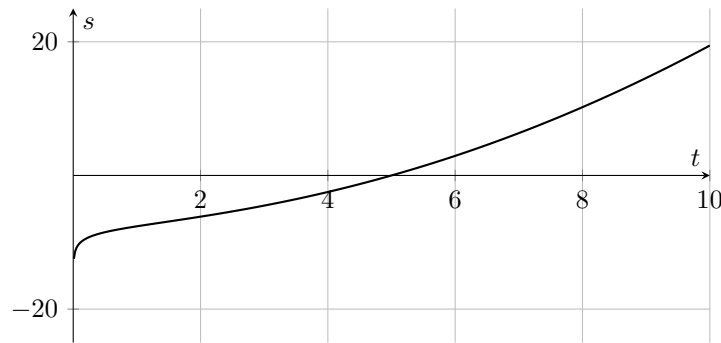
To obtain the arc length parameterization of a generic smoothly parameterized vector valued function $\vec{r}(t)$, first one computes the arc length as a function of t starting from a chosen reference point, say t_0 , by the formula

$$\vec{s}(t) = \int_{t_0}^t |\vec{r}'(t')| dt'$$

This gives s , the arc length, as a function of t , which you then solve for t in terms of s , and plug these new values for t in terms of s back into $\vec{r}(t)$ to get $\vec{r}(s)$

(b)

$$\begin{aligned} \vec{s}(t) &= \int_5^t \left(\frac{t'^2 + 2}{2t'} \right) dt' \\ &= \int_5^t \left(\frac{t'}{2} + \frac{1}{t'} \right) dt' \\ &= \left[\frac{t'^2}{4} + \ln(t') \right]_5^t \\ &= \frac{t^2}{4} + \ln(t) - \frac{25}{4} - \ln(5) \end{aligned}$$



(c)

For the change of parameters to be smooth we need $\frac{dt}{ds} \neq 0$ and $\frac{dt}{ds}$ to be continuous. We have

$$s = \frac{t^2}{4} + \ln(t) - \frac{25}{4} - \ln(5)$$

Then

$$\frac{ds}{dt} = \frac{t}{2} + \frac{1}{t}$$

$\frac{ds}{dt}$ is continuous and $\neq 0$, thus $s(t)$ is strictly monotonic and $\gamma^{-1}(t)$ is bijective, which implies $\frac{dt}{ds} \neq 0$ and $\frac{dt}{ds}$ continuous.

Thus $t = \gamma(s)$ is a smooth change of parameters